LAB 13

A/B-Test Stage: a Statistical Decision before Capital Moves

Item	Detail
Topic	Topic 3: Machine Learning Models, Statistical Modelling and A/B Testing for Trading Signals
Pipeline stage	A/B test
TSC reference	A5, K2
Duration	15 min
Folder	labs_fullcode/Lab13-ab-testing/
Files	Lab13-ab-testing/abtest_stage.py, pipeline/abtest.py, Lab13-ab-testing/solution/abtest.py
Prompts	Lab13_Prompts.pdf in this folder (step -> prompt, ready to copy)
Mode	Runs ONLINE with your Alpaca paper keys (.env), or OFFLINE on the bundled data and the paper-broker simulator. Paper trading only.

Lion City Capital and every scenario in these materials are fictitious and for training only. All trading in this course is paper (simulated) trading. Nothing in these materials is investment advice or a recommendation to trade any security.

Scenario

Two portfolio managers each champion a variant for the SPY sleeve: the 200-day trend filter and the 3-day dip buy. The pipeline's rule is that capital only moves between variants when an A/B test on the same days, with the same costs, says the difference is reliable - and the decision is recorded.

Objective

Build `python -m pipeline abtest` to compare any two library signals with a Welch t-test, a paired t-test and a block-bootstrap confidence interval for the Sharpe difference, and to record an accept or reject decision with a multiple-testing caution.

Before You Start

- Open a terminal in the labs_fullcode folder, activate your virtual environment and open your AI coding assistant there (set-up below). The complete, working code is already in this folder.
- Record every result you are asked to check - the same evidence is asked for in the Practical Performance assessment.

How to Run the Python Scripts

Everything runs from the labs_fullcode folder, ONLINE with your Alpaca paper keys or OFFLINE on the bundled data. Do the first four rows once; after that each lab is one command.

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Step	Command / file
Open a terminal in the labs_fullcode folder	cd labs_fullcode
Create and activate a virtual environment (once)	python3 -m venv .venv source .venv/bin/activate (Windows: .venv\Scripts\activate)
Install the libraries (once)	pip install -r requirements.txt
Optional: your Alpaca paper keys (leave blank to run OFFLINE)	cp .env.example .env (Windows: copy .env.example .env) then paste APCA_API_KEY_ID and APCA_API_SECRET_KEY into .env
Run Lab 13	python3 Lab13-ab-testing/abtest_stage.py (same as: python -m pipeline abtest)
The complete code	pipeline/abtest.py (reference copy: Lab13-ab-testing/solution/)
Outputs	_outputs/ (charts, CSV files and reports written by the stage)

Lab Steps - In Python with Your AI Assistant

AI writes the code; you own the algorithm. Paste each prompt into your AI coding assistant opened in the labs_fullcode folder (the same prompts, ready to copy, are in Lab13-ab-testing/Lab13_Prompts.pdf and Prompt_Library.html), run the stage and check the result. The complete, working code is already provided - see 'How to Run the Python Scripts' above. The Full Code Practical Performance uses this code path.

Step 1: State the hypotheses (A5, K2)

Write H0: both variants have the same expected daily return and Sharpe; H1: they differ. Choose alpha = 0.05 and note how many variants the desk has tried this quarter (default 20).

Check: H0, H1, alpha and the number of variants tried are written down before the test runs.

Step 2: Build the statistics (A5, K2)

Ask your assistant for the statistical core of pipeline/abtest.py.

AI prompt - paste into your AI coding assistant
Create labs_fullcode/pipeline/abtest.py. block_bootstrap_sharpe_diff(a, b, n_boot=2000, block=21, seed=7) resamples blocks of 21 consecutive days with numpy default_rng(seed) and returns Sharpe(A) - Sharpe(B) for each draw (annualised with sqrt(252)). ab_test(ra, rb, alpha=0.05, tried=20) returns the Welch t-test (scipy.stats.ttest_ind equal_var=False), the paired t-test (ttest_rel) and mean daily difference, the Sharpe difference and its (1 - alpha) bootstrap CI, the share of draws where A beats B, reject_h0 = paired p < alpha AND the CI excludes 0, and the Bonferroni threshold alpha / tried. Rules: run from the labs folder; reuse labs_fullcode/common (config, data, broker, indicators, metrics, backtest, plotting) and earlier pipeline stages instead of rewriting them; do not modify common/ or data/; read every setting from pipeline/config.yaml via pipeline.config.load_config(); never put API keys or secrets in code; paper trading only; keep a main(argv) function so `python -m pipeline <command>` works; add short comments a finance professional can follow.

Check: ab_test returns a dict with welch_p, paired_p, ci and reject_h0.

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Step 3: Build the A/B command (A5)

Ask your assistant for run() and main().

AI prompt - paste into your AI coding assistant
In labs_fullcode/pipeline/abtest.py add run(a='trend_filter', b='dip_buy', alpha=0.05, tried=20, symbol='SPY') for `python -m pipeline abtest --a --b --alpha --tried --symbol` (a and b chosen from pipeline.signals.SIGNALS): backtest both with the config cost model, use the SAME window from the first valid 200-day SMA, print CAGR, volatility, Sharpe, max drawdown, exposure and trades, the three tests, a DECISION line ('REJECT H0 - ...' or 'DO NOT REJECT H0 - <winner> looks better on paper, but the evidence is not reliable; keep both on paper') and the Bonferroni caution; write _outputs/Lab13_abtest_decision.yaml and chart Lab13_ab_test.png (both equity curves and the bootstrap histogram with the CI). Rules: run from the labs folder; reuse labs_fullcode/common (config, data, broker, indicators, metrics, backtest, plotting) and earlier pipeline stages instead of rewriting them; do not modify common/ or data/; read every setting from pipeline/config.yaml via pipeline.config.load_config(); never put API keys or secrets in code; paper trading only; keep a main(argv) function so `python -m pipeline <command>` works; add short comments a finance professional can follow.

Check: python -m pipeline abtest --help shows --a, --b, --alpha, --tried and --symbol.

Step 4: Run the A/B test (A5)

Run: python -m pipeline abtest --a trend_filter --b dip_buy --alpha 0.05 (same as: python3 Lab13-ab-testing/abtest_stage.py).

Check: Window 2018-10-16 to 2025-12-31 (1,812 days). A trend_filter vs B dip_buy: CAGR 9.70% vs 4.97%, Sharpe 0.81 vs 0.49, max drawdown -19.36% vs -23.93%, trades 24 vs 102.

Step 5: Compare the independent and paired tests (K2)

Read tests 1 and 2. Why does the paired design suit two strategies measured on the same days?

Check: Welch t = 0.74 (p = 0.462); paired t = 0.85 (p = 0.398), mean difference 0.00018 a day.

Step 6: Read the bootstrap CI and the decision (A5)

Read test 3 and the DECISION line, then explain the decision in plain words for the two portfolio managers.

AI prompt - paste into your AI coding assistant
Rewrite this A/B test result for two portfolio managers who are not statisticians, in under 100 words: <paste the three test lines and the DECISION line>. Explain why 'A looks better' is not the same as 'A is better', and what evidence would change the decision.

Check: Sharpe difference +0.31, 95% CI [-0.33, +1.05], A beats B in 82% of bootstrap samples; DECISION: DO NOT REJECT H0. Bonferroni with 20 variants: p < 0.0025.

Step 7: Test the pipeline's own strategy (A5, K2)

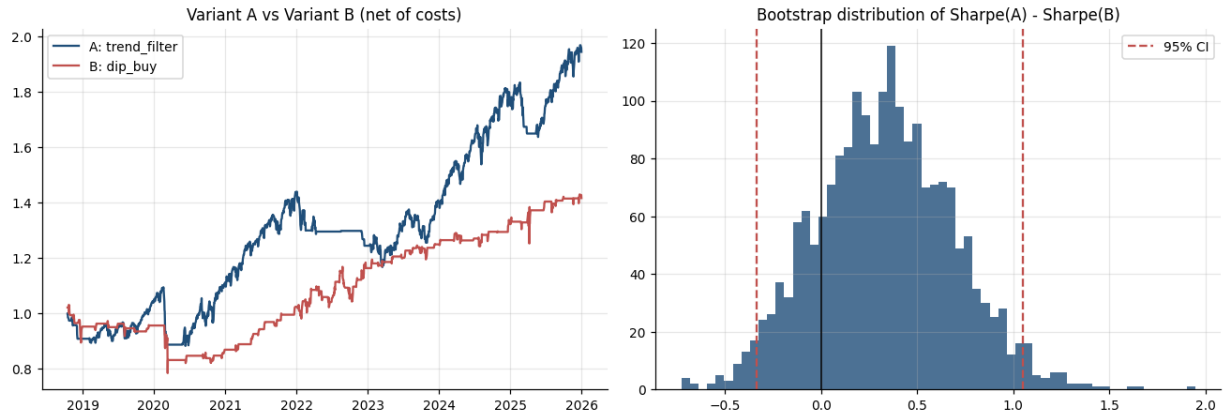
Run: python -m pipeline abtest --a ema_cross --b trend_filter.

Check: ema_cross Sharpe 0.95 vs trend_filter 0.81; Welch p 0.852, paired p 0.715, Sharpe difference +0.14, CI [-0.42, +0.67], A beats B in 66% - again DO NOT REJECT H0: the configured strategy is not proven better.

Expected Output (Full Code)

python -m pipeline abtest --a trend_filter --b dip_buy: window 2018-10-16 to 2025-12-31 (1,812 days); CAGR 9.70% vs 4.97%, Sharpe 0.81 vs 0.49; Welch t = 0.74 (p = 0.462); paired t = 0.85 (p = 0.398); Sharpe difference +0.31, 95% CI [-0.33, +1.05]; A beats B in 82% of bootstrap samples; DECISION: do not reject H0; Bonferroni p < 0.0025 for 20 variants; files _outputs/Lab13_abtest_decision.yaml and chart _outputs/Lab13_ab_test.png.

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Lab 13 output (labs_fullcode/_outputs/Lab13_ab_test.png), offline mode on the bundled data.

Debrief

A/B testing (A5) applies statistical inference to a business choice: the trend filter looks better, but the evidence cannot reject 'no difference'. Statistical tests (K2) - paired where the data are paired, bootstrapped where returns are not normal, corrected for the number of variants tried - protect capital from being moved on noise.

Troubleshooting

Symptom	What to do
p-value differs slightly	The bootstrap depends on the seed (7) and the number of draws; the decision should not change.
t-tests and the CI disagree	The t-tests compare mean returns, the CI compares Sharpe ratios; report both.
Returns are not normal	Daily returns have fat tails - that is why the block bootstrap is used alongside the t-tests.

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